Atharva Vidwans

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Ph.D. student making quantum algorithms practical by reducing the shots, queries, and circuit resources they need, with work spanning Bayesian amplitude estimation, Hamiltonian simulation, gate set tomography, and fault-tolerant resource estimation. Seeking a quantum computing research internship for Summer 2027.

Education

Ph.D. Chemistry, University of Wisconsin-Madison (expected May 2028) *Sep '24 – Present*
M.S. Physics (Quantum Computing), University of Wisconsin-Madison *Aug '22 – May '24*
B.E. Mechanical Engineering, Savitribai Phule Pune University *Jul '15 – Jun '19*

Publications

- **Atharva Vidwans** et al. "Not every circuit deserves a shot: adaptive shot experiment design for gate set tomography." Manuscript in preparation.
- *(co-first author)* Jingcheng Dai* and **Atharva Vidwans*** et al. "Molecular resonance identification in complex absorbing potentials via integrated quantum computing and high-throughput computing." arXiv preprint arXiv:2511.15981, 2025.
- *(co-first author)* Qilin Li* and **Atharva Vidwans*** et al. "Harnessing Bayesian Statistics to Accelerate Iterative Quantum Amplitude Estimation." *Quantum* **10**, 1962 (2026). arXiv:2507.23074.
- **Atharva Vidwans**. "Cognitive computing in autonomous vehicles." In *Cognitive Computing for Human-Robot Interaction*, pp. 121–146. Academic Press, 2021.

Work Experience

Argonne National Laboratory, Givens Associate (summer fellowship) *May '26 – Aug '26*
Advisors: Dr. Matt Menickelly and Dr. Jeffrey Larson

- Reduced required shots in gate set tomography by about 2.1x (one qubit) and 2.7x (two qubits) at the same gate accuracy, with no loss in SPAM accuracy, by allocating shots adaptively across circuits.
- Developed the allocation method using Fisher-information (D-optimal) experiment design, and designed a new criterion that fixes its SPAM-accuracy loss after showing with paired tests that the allocation, not the optimizer, caused it.
- Made the allocator 3.4x faster with an exact algorithm, bringing two-qubit runs inside the 72-hour cluster limit, and ran 800-run sweeps on HTCondor (CHTC) with pyGSTi and POUNDERS.
- Selected for the 2026 FASTMath Student Cohort (DOE SciDAC Institute) on mentor recommendation.

Research Experience

Bayesian and Resource-Estimation Methods for PDBQITE/DBQITE *Jan '25 – Present*
Advisor: Prof. Matthew Otten, UW-Madison

- Reduced queries per energy estimate by 4.2x (LiH) and 9.8x (BeH_2) in noiseless simulation, and a further 1.7 to 2.2x with an improved generator, using a Bayesian measurement scheme that reuses the failed ancilla outcomes PDBQITE throws away and picks each measurement to maximize information gain.
- Cut state-preparation cost by 4.1x (LiH) and 137x (BeH_2) with a recovery step that undoes a failed PDBQITE step instead of restarting, without changing the final state.
- Built fault-tolerant resource estimates with a surface-code wall-clock model (FLASQ) and double-factorized Trotter circuits compiled to Clifford+T (ffsim, Qiskit, WISQ), and ran systems up to Fe_2S_2 on HTCondor (CHTC).

Bayesian Iterative Quantum Amplitude Estimation (BIQAE) *Jan '23 – Jun '25*
Advisors: Prof. Micheline Soley and Prof. Yazhen Wang, UW-Madison

- Reduced sample complexity by about 14% compared to IQAE across six orders of magnitude in target accuracy, and by 8 to 13% for ground-state energies of H_2, LiH, BeH_2, and HF (co-first author, *Quantum* 2026).
- Built a unified statistical framework for amplitude estimation and added Bayesian inference to Iterative QAE, with rigorous intervals at every iteration.
- Showed with proofs and simulations that Bayesian inference alone accounts for the speedup.

qDRIVE for Molecular Resonances with Complex Absorbing Potentials *Jan '23 – Jun '25*
Advisor: Prof. Micheline Soley, UW-Madison

- Developed qDRIVE (co-first author), which combines the complex absorbing potential formalism with deflation-based VQE to find molecular resonance energies and wavefunctions.
- Ran the VQE task network asynchronously in parallel with HTCondor DAGMan to cut wall time.
- Recovered bound and resonance energies of a benchmark model to under 1% error in statevector simulation, and tested under shot noise, an IBM Torino noise model, and IBM hardware.

Surface-Code Logical Qubit Simulation *Feb '23 – May '23*
Advisor: Prof. Shimon Kolkowitz, UW-Madison

- Built X-cut and Z-cut surface-code logical qubits in Qiskit with logical X, Y, Z, and H gates, and started a braiding-based logical CNOT.
- Compared logical and physical qubits under gate and measurement noise up to 2%, and the logical qubits had 2 to 4.5x lower infidelity at the highest noise level.

Quantum Optimization for Vehicle Routing *Jun '21 – May '22*
Advisor: Prof. Pawel Gora, University of Warsaw

- Formulated the vehicle routing problem as a QUBO and implemented Layer-VQE in Qiskit, which grows the ansatz one identity-initialized layer at a time. Early experiments showed it outperforming standard VQE.
- Worked on Filtering VQE, a combined Layer-Filtering VQE, and CVaR objectives to improve accuracy.

Technical Skills

- **Quantum:** gate set tomography (pyGSTi), VQE, amplitude estimation (IQAE/BIQAE), Hamiltonian simulation, Trotterization, double factorization, surface codes, fault-tolerant resource estimation, Qiskit, ffsim, WISQ, OpenFermion, PySCF.
- **Statistics and optimization:** optimal experiment design, Fisher information, Bayesian statistics, derivative-free optimization (POUNDERS).
- **Computing:** Python, NumPy, SciPy, Numba, PyCUDA, HTCondor, Docker, Git, LaTeX.

Presentations

- "Efficient gate set tomography with POUNDERS." Talk, Argonne National Laboratory, Jul 2026.
- "Quantum Deflated Resonance Identification Variational Eigensolver (qDRIVE) Algorithm for Molecular Resonances on Near-Term Quantum Computers." **A. Vidwans**, J. Dai, E. Wan. Talk, Chicago Quantum Summit, 2024.
- "qDRIVE: A Variational Quantum Eigensolver for Resonance Identification on Near-Term Computers." Talk, Wisconsin Quantum Institute Colloquium, Mar 2024.
- "Quantum Deflated Resonance Identification Variational Eigensolver Algorithm for Molecular Resonances on Near-Term Quantum Computers." **A. Vidwans**, J. Dai. Poster, Graduate Student Faculty Liaison Committee, UW-Madison, Aug 2023.
- "Solving Vehicle Routing Problem Using Variational Quantum Eigensolver." **A. Vidwans**, W. el Maouaki. Talk, 13th Warsaw IT Days, Poland, Apr 2022.

Other Experience

Chegg India, Subject Matter Expert, Advanced Mathematics (part time) *Feb '21 – May '22*
Comofi Medtech Pvt. Ltd., Medical Imaging Intern *Feb '20 – Jul '20*
Advisor: Mr. Rakesh Sharma

- Reduced DRR generation time from 70 min to 8 sec for CT-to-X-ray registration in robotic kidney stone surgery, by writing a ray-casting algorithm in Python and parallelizing it with Numba. A PyCUDA kernel ran about 200x faster than Numba CUDA.
- Registered intra-operative C-arm X-rays against DRRs using MSE, SSIM, NCC, and feature-based homography to track kidney calyx shift for the robotic arm.

Awards, Service and Teaching

IBM Certified Associate Developer, Quantum Computation using Qiskit v0.2X	*Jan '22*
Rank 4 globally, IBM Qiskit Challenge 2021	*Jun '21*
Fourth position, National Level Qiskit Challenge, India (QWorld)	*Sep '20*
Ad hoc reviewer, *Quantum Machine Intelligence* (Springer)	*Jun '23*
Teaching Assistant, General Chemistry (CHEM 104, 109) and Physical Chemistry 2 (562)	*Sep '24 – Dec '25*